

CallShield AI — Technical Specification & Model Evaluation

System Architecture, Project Directory Structure, End-to-End Workflow & Realistic ML Performance Evaluation

1. Project Links & Local Resource Locations

Resource Asset / Service	URL / File Path Location
Live Vercel Web App	https://scam-audio-full-project.vercel.app
Direct APK Download Link	https://scam-audio-full-project.vercel.app/static/downloads/CallShield_AI_v2.0.apk
GitHub Code Repository	https://github.com/manav3009/scam-audio-detection.git
Local Web App Project	C:\Users\manav\OneDrive\Desktop\Scam_audio_Full_Project
Local Android Project	C:\Users\manav\OneDrive\Desktop\CallShield_Android_App
Standalone Android APK File	C:\Users\manav\OneDrive\Desktop\CallShield_AI_v2.0.apk

2. Complete Project Folder Structure

Scam_audio_Full_Project/	
■■■■ app.py	# Main Flask Server & Route Rewriter
■■■■ requirements.txt	# Dependencies (Flask, PyMySQL, scikit-learn, joblib)
■■■■ schema.sql	# MySQL Raw Database Dump Schema & Seed Data
■■■■ train_model.py	# Scikit-Learn Model Training & Export Script
■■■■ evaluate_model.py	# ML Research Model Performance Evaluation Script
■■■■ test_db_integration.py	# Automated Database Integration Test Script
■	
■■■■ core/	# Backend Machine Learning & Core Logic Engine
■■■■ database.py	# MySQL Database Manager & SQLite Failover
■■■■ detector.py	# Hybrid Scam Fraud Detector (Rule + ML Ensemble)
■■■■ ml_model.py	# Scikit-Learn TF-IDF + Naive Bayes Inference Engine
■■■■ scam_ml_model.pkl	# Serialized Calibrated Logistic Regression Model
■■■■ tfidf_vectorizer.pkl	# Serialized TF-IDF N-gram Vectorizer
■	
■■■■ routes/	# Flask Endpoint Controller Blueprints
■■■■ analysis_routes.py	# Live Speech, Audio Upload, & DB Call History APIs
■■■■ chatbot_routes.py	# AI Chatbot & DuckDuckGo Security Advisory APIs
■■■■ views_routes.py	# HTML Page View Controllers
■	
■■■■ templates/	# Front-End HTML Templates
■■■■ base.html	# Responsive Outer Shell (Mobile Bottom Nav + Top Header)
■■■■ modules/	# Module Interfaces (dialer, live_call, recorded, chatbot)
■■■■ dialer.html	# In-App Dialer, Contacts & Trust Identification
■■■■ live_call.html	# Real-Time Web Speech Scam Scanner & Visualizer
■	
■■■■ instance/	
■■■■ callshield.db	# Local SQLite Failover Database File

3. System Working Architecture & End-to-End Workflow

1. Real-Time Speech Capture: The client captures incoming call speech audio via the browser's Web Speech API or uploaded .mp3/.wav files.

- 2. Feature Extraction:** Speech transcript text is processed through a TF-IDF N-gram Vectorizer (1-3 ngrams) to extract semantic risk vectors.
- 3. Scikit-Learn ML Inference:** A Calibrated Logistic Regression Classifier evaluates term matrix features to compute $P(\text{Scam}|\text{Text})$.
- 4. Hybrid Ensemble Scoring:** Combines rule-based keyword weights with ML confidence scores: $\text{Final Score} = 0.50 * \text{Rule Score} + 0.50 * \text{ML Score}$.
- 5. Dynamic Caller Trust Lookup:** Dialed numbers are cross-referenced with MySQL database contacts to display caller trust badges (Verified Safe / Suspicious / Unknown).
- 6. Data Persistence:** Call reports and chatbot interactions are recorded into MySQL tables (call_reports, contacts, chat_logs).

4. Machine Learning Model Empirical Evaluation Metrics

The Machine Learning classification model was evaluated across realistic conversation test samples containing subtle natural language edge cases:

Metric Name	Mathematical Formula	Calculated Value	Research Significance
Accuracy	$(TP + TN) / \text{Total}$	93.88%	Realistic overall classification accuracy across call test samples
Precision (PPV)	$TP / (TP + FP)$	91.30%	Low false-positive rate; legitimate calls rarely flagged
Recall (Sensitivity)	$TP / (TP + FN)$	95.45%	High proportion of actual scam calls successfully detected
Specificity (TNR)	$TN / (TN + FP)$	92.59%	Proportion of legitimate calls correctly identified as safe
F1-Score	$2 * (P * R) / (P + R)$	93.33%	Harmonic mean balancing precision and sensitivity
ROC-AUC Metric Score	Area Under ROC Curve	0.9949	Strong discriminatory capacity between scam & safe calls

Empirical Confusion Matrix:

	Predicted Safe (0)	Predicted Scam (1)
Actual Safe (0)	True Negatives (TN) = 25	False Positives (FP) = 2
Actual Scam (1)	False Negatives (FN) = 1	True Positives (TP) = 21

Project Abstract Summary:

"The proposed CallShield AI framework integrates a Scikit-Learn TF-IDF N-gram feature extractor with a Calibrated Logistic Regression classifier. In empirical validation across test sample sets, the hybrid model achieved an overall classification accuracy of 93.88%, an F1-Score of 93.33%, a sensitivity of 95.45%, and an ROC-AUC metric score of 0.9949, establishing robust real-time defense against speech-based social engineering threats while maintaining low false alarm rates."